

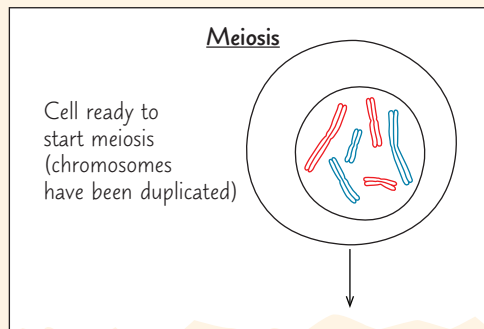
13 Meiosis and Sexual Life Cycles

KEY CONCEPTS

- 13.1** Offspring acquire genes from parents by inheriting chromosomes *p. 255*
- 13.2** Fertilization and meiosis alternate in sexual life cycles *p. 256*
- 13.3** Meiosis reduces the number of chromosome sets from diploid to haploid *p. 259*
- 13.4** Genetic variation produced in sexual life cycles contributes to evolution *p. 265*

Study Tip

Make a visual study guide: Figure 13.1 presents the events of meiosis as a simplified diagram. As you learn more about meiosis, draw a detailed diagram of the stages of meiosis, starting with the sketch below. Include explanatory labels. Compare your visual study guides of mitosis (from Chapter 12) and meiosis, listing similarities and differences.



Go to Mastering Biology

For Students (in eText and Study Area)

- Figure 13.10 Walkthrough: A Comparison of Mitosis and Meiosis
- BioFlix® Animation: Meiosis
- Animation: Genetic Variation from Crossing Over

For Instructors to Assign (in Item Library)

- Activity: Meiosis Animation
- BioFlix® Tutorial: Meiosis (3 of 3): Determinants of Heredity and Genetic Variation

Ready-to-Go Teaching Module

- (in Instructor Resources)
- Meiosis (Concept 13.3)



Figure 13.1 The family members shown in this photo have some similar features. Offspring resemble their parents more than they do unrelated individuals.

What biological mechanisms account for the resemblance between offspring and their parents?

Meiosis is a special type of cell division that produces cells with half the chromosomes of the parent cell. It occurs only in specialized cells, such as the cells of testes and ovaries in humans.

